

Used Car Price Estimator

Task 04 — End-to-End ML Application with Deployment | AI/ML Internship, Devixo Solutions
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Objective

Predict the resale price of a used car in Pakistan from listing details (brand, age, mileage, fuel type, city, sale type), and serve the prediction through a live web app rather than a notebook.

Dataset

OLX Pakistan used-car listings — ~25,000 raw rows, cleaned down to 15,650. Cleaning handled missing values, ~3,800 duplicates, mixed new/used listings (filtered to used only), a mislabeled brand entry, and outlier prices/mileage/year trimmed with a widened IQR rule.

Approach

- Feature engineering: car age from year, rare brands/cities bucketed into "Other".
- Trained and compared 4 regressors: Linear Regression, Decision Tree, Random Forest, Gradient Boosting.
- Tuned the best model (Gradient Boosting) with RandomizedSearchCV (5-fold CV, 25 iterations).
- Saved the full preprocessing + model pipeline with joblib.
- Built a Streamlit app for input → prediction → result, deployed on Streamlit Community Cloud.

Model Comparison

Model	MAE (Rs)	RMSE (Rs)	R ²
Linear Regression	2,00,068	2,89,925	0.691
Decision Tree	1,72,799	2,96,663	0.676
Random Forest	1,51,231	2,50,804	0.768
Gradient Boosting	1,49,463	2,45,305	0.779
Gradient Boosting (tuned)	1,43,287	2,38,830	0.790

Best params: n_estimators=200, max_depth=5, learning_rate=0.05, subsample=0.7. Trained on 15,650 listings, evaluated on a held-out 20% test split.

Result & Deployment

Final model: **Gradient Boosting (tuned)** · R² = **0.79** · RMSE ≈ **Rs 2,38,830** on unseen listings.

Live app: anayaafarhan-ml4-app-azsuiz.streamlit.app

Code: github.com/anayaafarhan/ml4